

ZIYAO SU

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EDUCATION

The Ohio State University, *Ph.D.* in Materials Science and Engineering | GPA: 3.96/4.00 2023 – 2027
The Ohio State University, *M.S.* in Materials Science and Engineering | GPA: 3.94/4.00 2023 – 2024
The Ohio State University, *B.S.* in Physics and Mathematics, Minor in Music | GPA: 3.73/4.00 2018 – 2022

AWARDS AND HONORS

- MJ Soileau Best Student Paper Award (2025 SPIE Laser Damage Conference)
- The Ohio State University Undergraduate Dean's List (6 semesters: Fall 2018, Fall 2019, Spring 2020, Spring 2021, Fall 2021, Spring 2022)

TECHNICAL SKILLS

- **Personal Ability:** Independently developed multiphysics laser-matter interaction code; familiar with laser experiments.
- **Computational Methods:** FDTD, PIC (EPOCH), RCWA(RETICOLO), DFT (VASP)
- **Programming Languages:** Python, Matlab, C++, Java
- **Software Tools:** Mathematica, VisIt, CrystalMaker, SNLO

RESEARCH EXPERIENCE

Graduate Research Assistant, Femto-Solid Laboratory, OSU Sep 2021 – Present

- ★ Research focus: FDTD and PIC modeling of ultrafast laser-matter interactions and LIDT analysis.
- Independently developed 2D and 3D FDTD simulations in MATLAB to investigate femtosecond laser interactions and damage thresholds in metallic and dielectric multilayer mirrors, gratings, and nanolaminate thin films. Implemented a Total-Field/Scattered-Field (TFSF) source and integrated multiple physical models, including Drude response, Keldysh hybrid ionization, collisional ionization, and electron-collision processes.
- Developing a hybrid simulation framework combining in-house FDTD and open-source RCWA methods for optimizing 2 μm grating designs, alongside Bayesian machine learning models to further optimize the designs.
- Conducted PIC simulations via EPOCH to study nonlinear electron dynamics and energy spectra in petawatt laser interactions with multi-phase targets (water films/ $C_2H_6O_2$ films/Ta foils); developed Keldysh ionization modeling in EPOCH for accurate simulation of femtosecond laser-solid multilayer mirror interactions and damage threshold prediction, and executed large-scale simulations on HPC platforms (Unity Cluster, OSC, NERSC) with optimized MPI/OpenMP parallelization.
- Collaborated with Colorado State University and XUV Lasers Inc. to validate laser damage thresholds of various optical components (mirrors and gratings).

SELECTED PUBLICATIONS

Journal Articles

- **Ziyao Su**, and Enam A. Chowdhury. "Pulse-Duration Scaling of Laser-Induced Damage Threshold in Hybrid Gratings under Ultrafast Irradiation." *Optics Letters Opt. Lett.* 51, 4236-4239 (2026).
- **Ziyao Su**, and Enam A. Chowdhury. "Ultrafast Laser Interaction with Dielectric-Metal Hybrid Coatings: A 2D FDTD Modeling Study" *High Power Laser Science and Engineering*, pp. 1–17, 2026.
- Joseph R. Smith, **Ziyao Su**, Simin Zhang, Charles Varin, Vitaly E. Gruzdev, and Enam A. Chowdhury. "A Fully Three-Dimensional Kinetic Particle-In-Cell Framework for Modeling Laser-Dielectric Interactions: Few-Cycle Pulse Damage." *High Power Laser Science and Engineering*. Published online 2025:1-16. doi:10.1017/hpl.2025.10041
- Simin Zhang, **Ziyao Su**, Carmen S. Menoni, and Enam A. Chowdhury. "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings." *Optics Letters* 48, 1212–1215 (2023).
- Milo Eder, Justin Twardowski, **Ziyao Su**, Alphonse Marra, Michael Chini, Zenghu Chang, and Enam Chowdhury. "X-ray production from high intensity mid-IR laser interactions with thin liquid sheet target in air." *Ultrafast Science*, **Under Review**, 2026.

Conference Proceedings

- **Ziyao Su**, and Enam A. Chowdhury. "FDTD Modeling of Femtosecond Laser-Hybrid Grating Interaction and Damage Mechanisms." In *CLEO 2026, Technical Digest Series (Optica Publishing Group, 2026)*, paper JW1.123.
- **Ziyao Su**, Mohamed Yaseen Noor, Emma DeAngelis, Carmen S. Menoni, Enam A. Chowdhury. "Keldysh ionization based-FDTD modeling of laser-induced damage threshold of MLD-IBS compression gratings for petawatt 2 μm laser systems." *Proc. SPIE PC13729, Laser-Induced Damage in Optical Materials 2025, PC1372907 (19 December 2025)*.

- **Ziyao Su**, Joseph R. Smith, and Enam A. Chowdhury. "Electron dynamics of femtosecond laser induced plasma inside multilayer dielectric high reflectors studied using particle-in-cell (PIC) approach." In *CLEO 2024, Technical Digest Series (Optica Publishing Group, 2024)*, paper SF3H.5.

Presentations

- **Ziyao Su**, Enam A. Chowdhury. "FDTD Modeling of Sub-Picosecond Laser Interaction and Damage in Hybrid Optical Coatings." poster presented at *Proc. SPIE 14074-47, High-Power Laser Ablation IX, 14074-47X 2026, (March 2026)*.
- **Invited Talk: Ziyao Su**, and Enam A. Chowdhury. "2D FDTD Modeling of Ultrashort Laser-Matter Interactions," presented virtually at *Laboratory for Laser Energetics (LLE), University of Rochester*, November 5, 2025.
- **Ziyao Su**, Joseph R. Smith, and Enam A. Chowdhury. "Electron Dynamics of Femtosecond Laser Induced Plasma Inside Multilayer Dielectric High Reflectors Studied Using a Particle-in-Cell (PIC) Approach," poster presented at *LaserNetUS User Meeting, 2024*.

ACADEMIC SERVICE AND SOCIETY MEMBERSHIPS

- Journal Reviewer (8): *Applied Optics* (2024, 2025), *Optics Express* (2026), *Optics Letters* (2026).
- Society Memberships: APS (2024, 2026); SPIE (2024-2026). SPIE OSU Chapter (2024-2026)

TEACHING EXPERIENCE

Instructional Assistant, MSE 2010 Introduction to Engineering Materials Sep – Dec 2024

- Facilitated instruction on structure–property–processing relationships across metals, ceramics, polymers, and composites.
- Led problem-solving sessions and provided assignment support, reinforcing understanding of mechanical, thermal, and electrical material properties.

Instructional Assistant, MSE 3331 Materials Science and Engineering Laboratory I Sep – Dec 2025

- Supported laboratory modules on steel thermomechanical processing and aluminum alloy design, emphasizing microstructure evolution, heat treatment, corrosion behavior, and performance evaluation.
- Guided student teams in microstructure characterization and technical report writing; evaluated laboratory submissions and provided feedback.

ACTIVITIES & LEADERSHIP

- Member of Chinese Students and Scholars Society (CSSS) and Puzzleology Club – Event Planning and Publicity (Sep 2024 – Present)
- Registered USATT Player, Columbus Table Tennis Club; Varsity team member, represented The Ohio State University in NCTTA competitions; USATT Rating: 1953 (Jun 2025 – Present)